

Project Title: How Do People Read Dashboards?
A Dashboard Interpretation and Decision-Making Study

An exploratory study evaluating how users interpret operational dashboards and translate insights into business decisions using survey responses, Python, and qualitative analysis.

Project Overview

Dashboards are designed to transform data into actionable insights. However, a dashboard is only effective if users can correctly interpret the information it presents and make informed decisions from it.

This study evaluated how users from different backgrounds interpreted a Logistics Operations Performance Dashboard. Rather than assessing the dashboard solely on its visual appearance, the study explored how users interacted with it, what attracted their attention, how they interpreted the information, the operational concerns they identified, and the actions they prioritized based on the insights presented.

The goal was to understand whether the dashboard effectively communicated information and supported decision-making while identifying opportunities for improving dashboard design and storytelling.

Project Resources

Google Colab Notebook: [How Do People Read Dashboards?](#)

Google Forms Questionnaire: [Dashboard Interpretation Survey](#)

Business Problem

Organizations invest significant time building dashboards, but dashboards are only valuable if users can quickly understand the information being presented.

Dashboards that are difficult to interpret may lead to:

- Misinterpretation of business information
- Poor decision-making
- Increased cognitive effort
- Important insights being overlooked

This study sought to understand how users naturally interact with a dashboard and identify design elements that improve interpretation and decision-making.

Objective of the Study

The objectives of this study were to:

- Evaluate how effectively the dashboard communicates its intended purpose.
- Identify the dashboard elements that attract users' attention first.
- Determine which dashboard sections users spend the most time interpreting.
- Explore the operational concerns users identify from the dashboard.
- Understand the actions users would prioritize after viewing the dashboard.
- Gather user feedback to improve future dashboard design.

Research Questions

The study was guided by the following questions:

- What catches users' attention first?
- What do users believe the dashboard represents?
- Which dashboard section receives the most attention?
- What operational concern would users investigate first?
- What action would users prioritize?
- Why did users spend more time on particular dashboard sections?
- What improvements can be made to enhance dashboard communication?

Dashboard Used in the Study

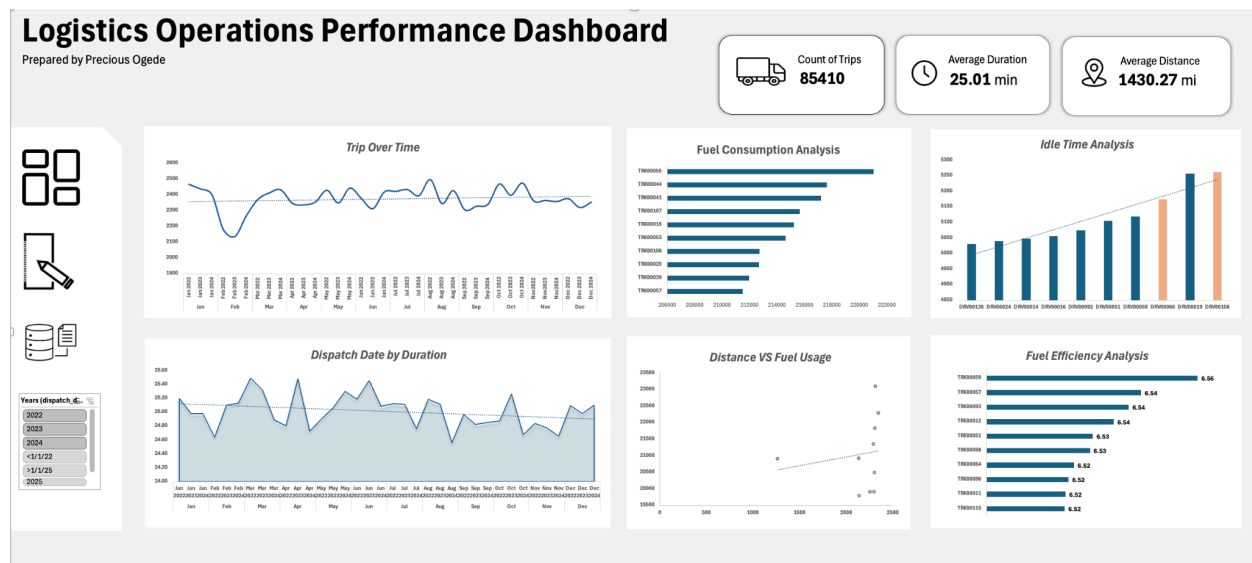


Figure 1: Dashboard presented to participants before completing the survey.

Methodology

A structured questionnaire was created using **Google Forms** and distributed to participants.

Participants were instructed to carefully examine the dashboard before answering a series of quantitative and qualitative questions.

The survey collected both categorical responses and open-ended feedback to provide a balanced understanding of dashboard interpretation.

Participants

A total of **13 participants** took part in the study. Participants represented a diverse range of professional backgrounds, allowing the dashboard to be evaluated from multiple perspectives rather than a single area of expertise.

The participants' professions included:

- Accountant
- Customer Experience Professional
- Data Analyst

- Engineer
- Lawyer
- Marketing Professional
- Product Designer
- Project Manager
- Quality Assurance (QA) Professional
- Teacher
- Web Developer

Note: Some professional backgrounds had more than one participant, while others were represented by a single participant. The diversity of the sample was intended to provide varied perspectives on dashboard interpretation rather than to compare responses across professions.

Tools Used

Dashboard Development

- Microsoft Excel

Data Collection

- Google Forms

Data Analysis

- Python
- Pandas
- NumPy

Data Visualization

- Matplotlib

Development Environment

- Google Colab

Data Preparation

The survey responses were exported from Google Forms as a CSV file and imported into Google Colab.

The dataset was reviewed to ensure it was suitable for analysis.

The following preprocessing steps were performed:

- Imported the dataset using Pandas
- Examined the structure of the dataset
- Reviewed column names and data types
- Checked for missing values
- Standardized categorical responses where necessary
- Prepared frequency distributions
- Generated visualizations using Matplotlib

Exploratory Data Analysis (EDA)

The analysis focused on understanding participant responses across several key questions.

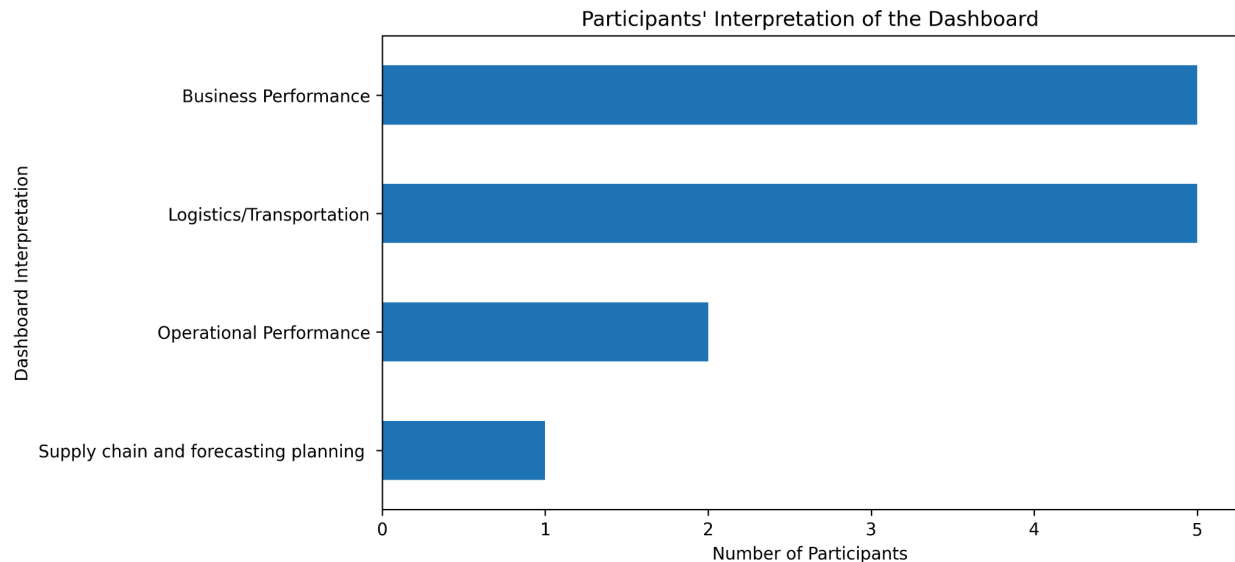
The following visualizations were created:

- Dashboard interpretation
- Elements that first attracted attention
- Sections participants spent the most time examining
- Operational concerns identified
- Actions participants would prioritize

Open-ended responses were also reviewed and grouped into common themes.

Findings

Finding 1: Participants correctly interpreted the dashboard's purpose



Observation

Most participants correctly identified the dashboard as relating to logistics, transportation, or business performance.

Although participants described the dashboard differently, their responses consistently reflected its intended purpose.

Interpretation

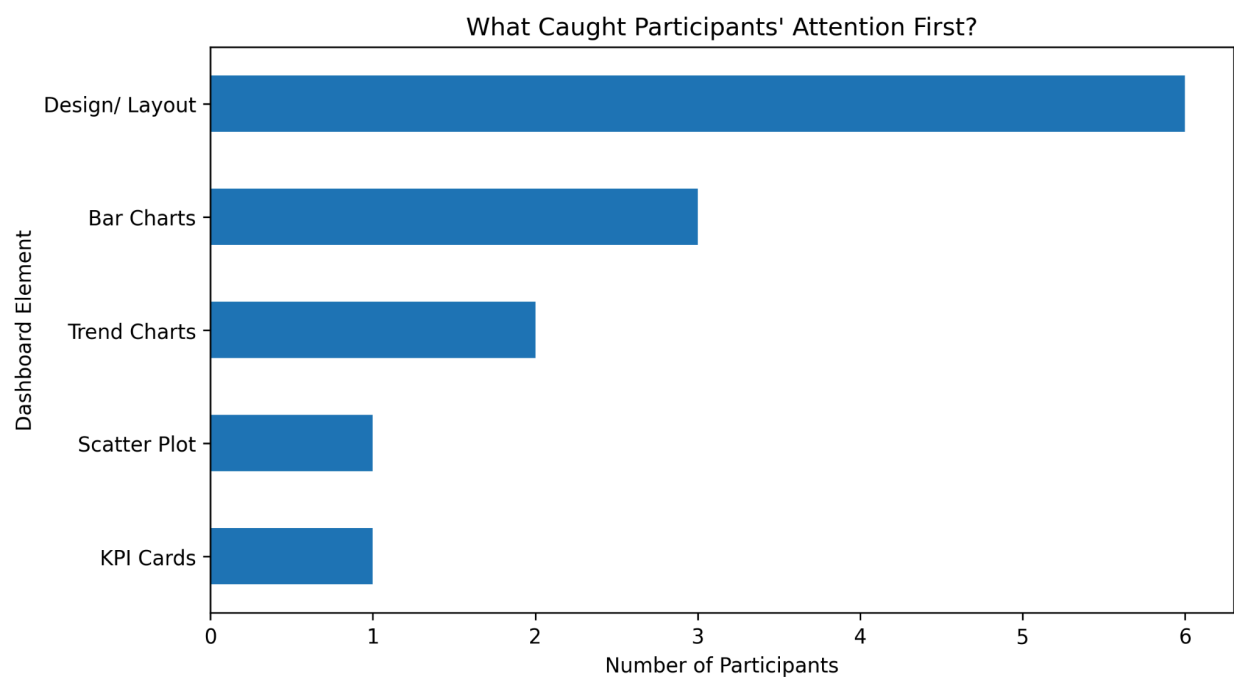
This suggests that the dashboard effectively communicates its objective without requiring additional explanation.

The dashboard title, KPI cards, and operational charts provided enough context for users to understand what they were viewing.

Implication

An effective dashboard should reduce the amount of time users spend trying to understand its purpose, allowing them to focus on interpreting insights instead.

Finding 2: Summary information attracted users' initial attention



Observation

The responses indicate that many participants first noticed the dashboard's high-level summary information, including KPI cards and other prominent visual elements. These elements appeared to attract initial attention before participants explored other sections of the dashboard. This observation suggests that users often begin by scanning the dashboard to gain a general understanding before examining more detailed information.

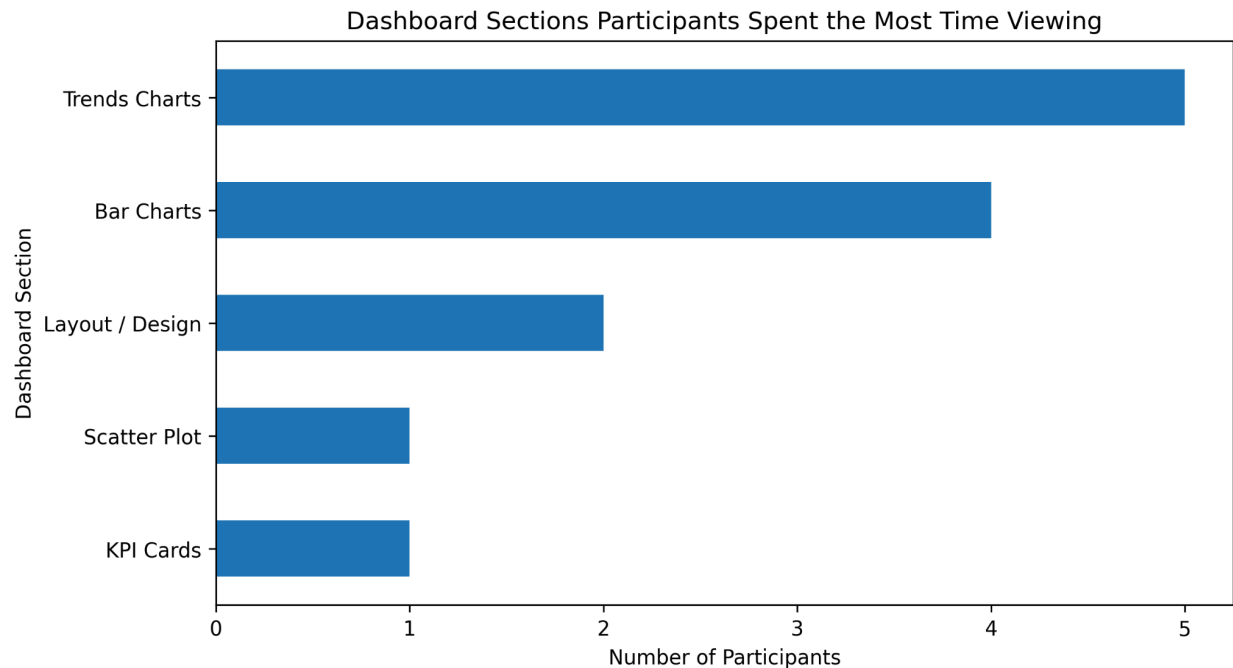
Interpretation

The pattern observed highlights the role of **visual hierarchy** in dashboard design. Elements that are larger, more prominently positioned, or visually distinct naturally draw attention and help users establish an initial understanding of the information being presented. Summary metrics appear to provide a starting point from which users begin exploring the rest of the dashboard.

Implication

When designing dashboards, it is beneficial to position important summary information where users are likely to notice it early. A clear visual hierarchy can help users understand the dashboard more quickly and provide context before they begin interpreting detailed charts and operational metrics.

Finding 3: Users spent more time analyzing detailed visualizations



Observation

Although many participants initially noticed the dashboard's summary information, their responses indicate that they spent more time examining the detailed charts and visualizations.

The qualitative responses also revealed several reasons for this. Participants explained that they spent more time on particular charts because they:

- wanted to better understand the statistics,
- found the charts easy to interpret,
- considered the visualizations well organized,
- found the charts visually appealing, or
- believed the charts communicated information clearly.

Interpretation

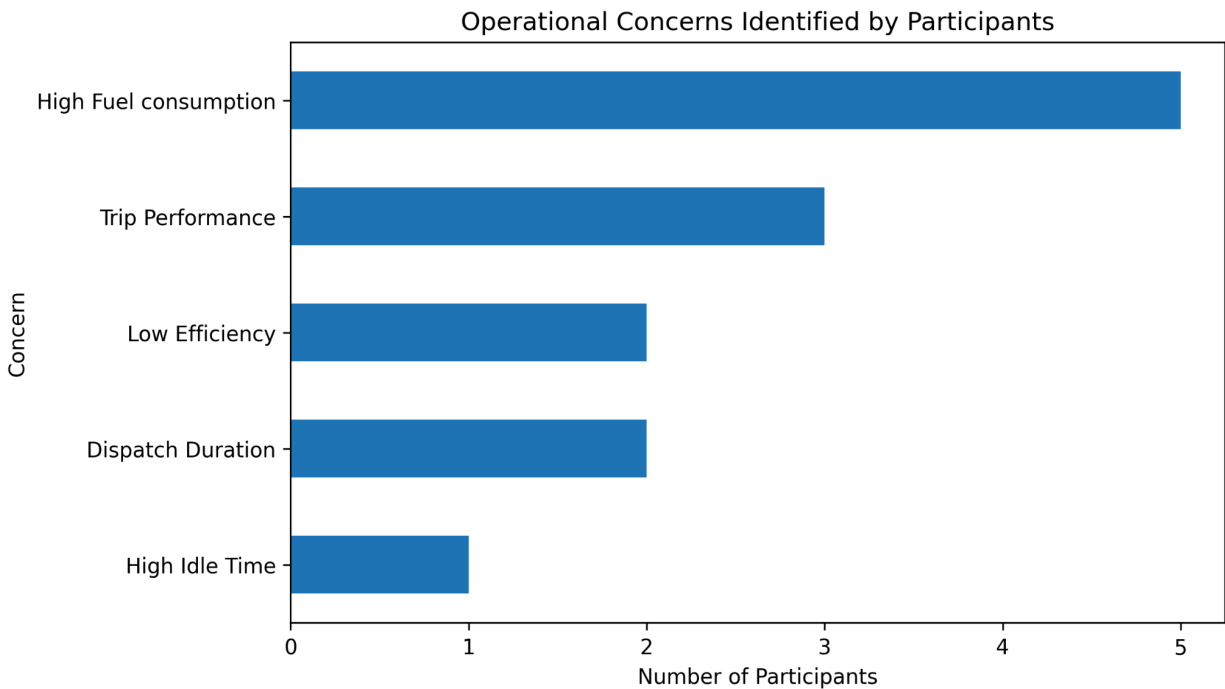
The findings suggest that while summary metrics may capture initial attention, detailed visualizations support deeper exploration and understanding.

Participants appeared to spend additional time on charts that helped them interpret trends and understand the operational information presented in the dashboard.

Implication

A dashboard should not rely solely on summary metrics. While summary information provides an overview, detailed visualizations play an important role in helping users explore the data, understand patterns, and make informed decisions. Combining both elements can create a more effective and user-friendly dashboard.

Finding 4: Participants identified meaningful operational concerns



Observation

When asked which operational issue would concern them most, participants selected different areas of the dashboard. Their responses included fuel consumption, idle time, dispatch duration, trip performance, and overall operational efficiency.

The variation in responses suggests that participants did not focus on a single metric. Instead, they prioritized different operational concerns based on what they considered most important.

Interpretation

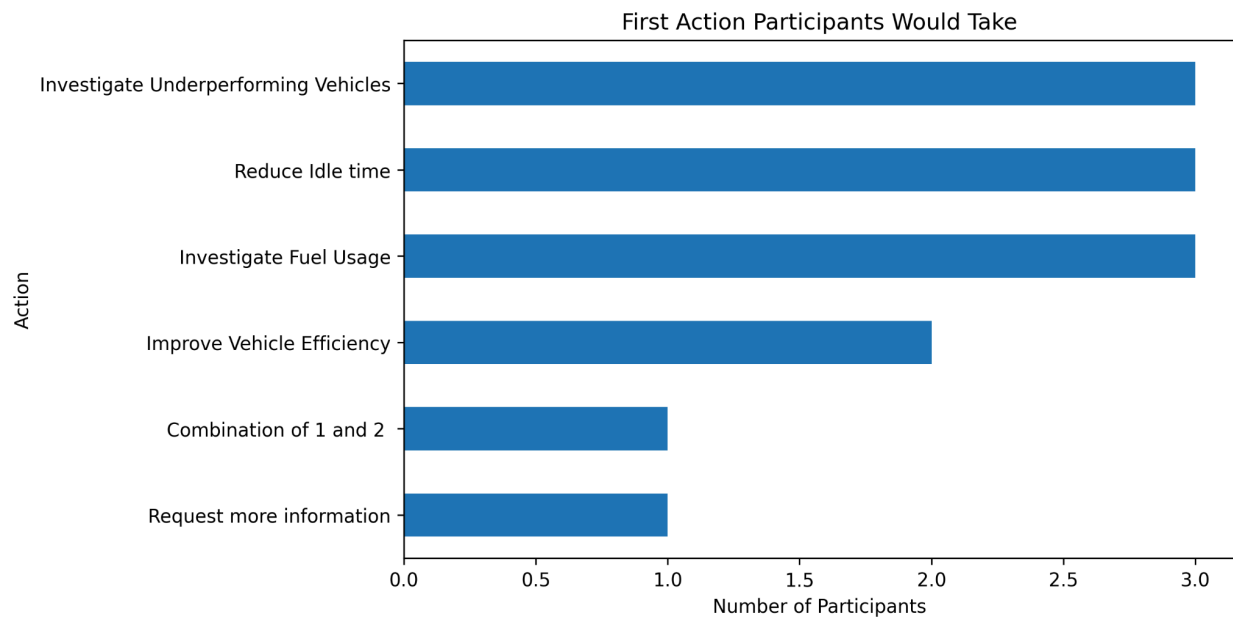
Rather than simply describing what they observed, participants evaluated the dashboard from an operational perspective and identified the issue they believed required the most immediate attention.

This suggests that participants interpreted the information presented in the dashboard and applied their own judgment to determine which operational concern they would prioritize.

Implication

The findings demonstrate that dashboards can support different perspectives depending on the user's priorities and decision-making context. Presenting multiple operational metrics allows users to identify the areas they consider most critical and supports more informed business discussions and decision-making.

Finding 5: The dashboard encouraged action-oriented thinking



Observation

The actions participants chose closely reflected the operational concerns they identified earlier. Participants generally selected a first course of action that addressed the issue they considered most important, such as investigating fuel consumption, reducing idle time, improving dispatch efficiency, reviewing trip performance, or monitoring overall operational performance.

This suggests that participants linked the insights they observed in the dashboard to practical actions they believed would have the greatest operational impact.

Interpretation

Rather than responding with generic recommendations, participants used the information presented in the dashboard to determine what they would address first. Their responses indicate that different participants prioritized different actions based on the operational concern they identified as most significant.

This suggests that participants were able to interpret the dashboard, assess the information from their own perspective, and translate those insights into practical business actions.

Implication

The findings highlight that dashboards support decision-making by providing information that users can interpret according to their priorities and responsibilities. When operational metrics are presented clearly, users are better equipped to identify the issue they consider most important and determine an appropriate first course of action.

Qualitative Insights

The open-ended responses provided additional insight into how users interacted with the dashboard.

Several recurring themes emerged:

Theme	Participant Feedback
Ease of interpretation	"It was easy to interpret."
Visual appeal	"It caught my attention."
Information richness	"I wanted to read the statistics."
Simplicity	"The bar chart was straightforward."
Organization	"It looked well arranged."

Interpretation

Participants consistently spent more time on charts they perceived as clear, informative, and visually organized.

This suggests that readability and presentation significantly influence how users engage with dashboard visualizations.

Key Takeaways

This study highlighted several important lessons about dashboard design and user interaction.

1. Dashboard titles and summaries shape first impressions.

Users first looked for information that helped them quickly understand the dashboard before exploring detailed charts.

2. Clear visualizations encourage deeper engagement.

Participants spent more time on charts that were visually organized, easy to understand, and information-rich.

3. Effective dashboards support decision-making.

Participants were able to identify operational issues and recommend actions without additional explanation.

This demonstrates that dashboards should facilitate decision-making rather than simply display data.

4. Simplicity improves communication.

Participants frequently described the charts they focused on as clear, straightforward, and easy to interpret.

This reinforces the importance of designing dashboards that prioritize clarity over complexity.

5. User feedback is valuable for dashboard improvement.

Collecting feedback from dashboard users provides valuable insights into how dashboards communicate information and where design improvements can be made.

Limitations

This study has several limitations:

- The sample size was relatively small.
- Participants had different levels of analytical experience.
- Responses represent individual interpretations and should not be generalized to all dashboard users.

Despite these limitations, the study provides useful insights into dashboard usability and interpretation.

Lessons Learned

This study reinforced several important principles of effective dashboard design and evaluation.

1. A dashboard should communicate before it impresses.

The findings showed that participants were generally able to understand the purpose of the dashboard without additional explanation. This highlights the importance of clear titles, meaningful KPIs, and intuitive visual design. A dashboard's first responsibility is to communicate information effectively.

2. First attention is different from sustained attention.

While participants were initially drawn to summary metrics and prominent dashboard elements, they spent more time examining detailed visualizations before forming conclusions. This demonstrates that dashboards should balance high-level summaries with supporting visuals that provide context and encourage deeper analysis.

3. Simplicity improves user engagement.

Participants frequently described the charts they focused on as easy to interpret, well organized, and visually appealing. This reinforces the idea that effective dashboards do not need to be overly complex. Clear layouts, readable charts, and thoughtful visual hierarchy significantly improve the user experience.

4. Effective dashboards support decision-making.

Rather than simply describing the information displayed, participants identified operational concerns and proposed practical actions they would take. This suggests that a well-designed dashboard should guide users beyond observation and support informed, data-driven decision-making.

5. User feedback is an essential part of dashboard development.

Building a dashboard is only one part of the process. Evaluating how users interact with it provides valuable insights into how effectively it communicates information. Gathering feedback helps identify strengths, uncover areas for improvement, and ensure the dashboard meets the needs of its intended audience.

6. Dashboard evaluation goes beyond technical implementation.

This project demonstrated that dashboard development is not only about creating visualizations but also about understanding how users interpret information. Combining quantitative analysis with qualitative feedback provided a more complete evaluation of the dashboard's effectiveness and highlighted the value of user-centered design.

Conclusion

This project demonstrates that evaluating a dashboard extends beyond checking whether charts and KPIs are visually appealing.

A successful dashboard should communicate its purpose clearly, guide users toward meaningful insights, and support informed decision-making.

By combining survey data with exploratory data analysis and qualitative feedback, this study evaluated not only how the dashboard looked, but also how users interpreted and interacted with it.

The findings reinforce the importance of user-centered dashboard design and show that effective visualizations should reduce cognitive effort while encouraging analytical thinking and actionable decisions.

Technical & Analytical Skills Demonstrated

Technical

- Python
- Pandas
- NumPy
- Matplotlib
- Google Colab
- Excel

Analytical

- Survey Design
- Dashboard Evaluation
- Exploratory Data Analysis
- Qualitative Analysis
- Business Interpretation
- Data Storytelling
- Insight Generation